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V300 ZEFIRO

A New Sense of Speed

Interior Lighting- VF.CONCEPT SFRS

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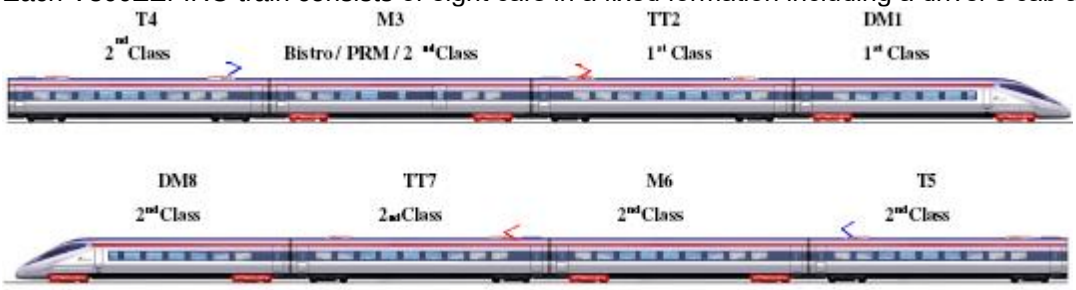
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1 Introduction

This document is a function specification for the control of the Internal Lighting system for V300ZEFIRO Spain for customer Trenitalia.
The lighting system of the electronic room and drivers' cab **is not** in the scope of this document.

1.1 Train Configuration

Each V300ZEFIRO train consists of eight cars in a fixed formation including a driver's cab at each end:



- DM = End driving motor car with driver's cab
- TT = Intermediate trailer car with transformer
- M = Intermediate driving motor car
- T = Intermediate trailer car

1.2 Definitions and abbreviations

CD	Circuit Diagram
EVENT	Diagnostics that will be stored in the Event database within TCMS SW.
HW HMI	Hardware Human Machine Interface, e.g. buttons, switches and indication lamps.
IDU	Interface Display Unit
TCMS	Train Control and Management System
LV1	Normal lighting power supply line 1
LV3	Normal lighting power supply line 2
LV2-1	Emergency lighting power supply line 1
LV2-2	Emergency lighting power supply line 2
LV5	Aesthetical & vestibule lighting power supply line
LV4 -1	Safety lamp power supply line
LV4 -2	Safety lamp switch on/off command (train level)
LV4 -3	Safety lamp 1 - battery status
LV4 -4	Safety lamp 2 - battery status
TL	Train line

2 System Overview

2.1 System Configuration

The philosophy and the architecture of the interior lighting system of each coach is the same for the whole train.
In each coach the following lighting areas can be identified:

- passenger compartment and vestibule lighting (general lighting)
- toilet lighting
- bistro lighting (M3 coach only)
- cabin crew lighting (M3 coach only)

Passenger compartment and vestibules lighting (general lighting)
Lighting system is composed by several LED spot lights installed on the ceiling of the vestibules and passenger compartment. Its aim is to fulfil the lighting performances in terms of lighting level and uniformity according to the applicable standard (EN13272 and UIC555) during normal and emergency condition.
A part of main lighting, controlled by hardwired circuitries, provides the function “emergency lighting”. It includes the access steps lights and the gangway lights.
General lighting is divided in 5 power supply lines:

- Saloon lighting power supply line 1 (LV1)
- Saloon lighting power supply line 2 (LV3)
- Emergency lighting power supply line 1 (LV2-1)
- Emergency lighting power supply line 2 (LV2-2)
- Aesthetical & vestibules lighting power supply line (LV5)

In normal condition (e.g. battery charger correctly working) General Lighting is controlled in the same way (switch on and off together) so, when a general lighting switch on command is issued all the above power supply lines will be activated:

- Saloon lighting power supply line 1 (LV1) (if the full light mode is selected)
- Saloon lighting power supply line 2 (LV3) (if the full light mode is selected)
- Emergency lighting power supply line 1 (LV2-1)
- Emergency lighting power supply line 2 (LV2-2)
- Aesthetical & vestibules lighting power supply line (LV5)

The switching on/off command devices are available in the driver cabs, in the crew cabin and in all coaches. In addition to that, also the TCMS can control the lighting system in some particular condition (i.e. desk activation, tunnel mode..).
From the crew cabin monitor it is possible to set the full light or reduced light mode. When in a coach the reduced lighting mode is selected, in that coach LV1 and LV3 will remain switched off.
The TCMS can not disconnect the emergency lighting lines. Only the staff can switch off the emergency lighting lines acting on the command devices present in the driver cabs and in the crew cabin
In case of failure of the battery chargers (train powered by the battery system), in order to contain the batteries discharging an automatic disconnection of the lighting load is performed by the TCMS. At the end of this automatic sequence, only the emergency lighting will remain active.

Toilet lighting
Toilet lights are divided in two independent power supply lines. One of these is connected directly to the toilet power supply line so, when the toilet is active, these lights are switched on. The second one is connected to the emergency lighting of the train.

Cabin crew and bistrot lighting
Both Cabin crew and bistrot lighting systems are controllable locally by the staff by means of dedicated control devices.

3 Functional requirements

3.1 Interior Lighting

3.1.1 General Lighting

3.1.1.1 Design description

In normal condition (e.g. battery charger correctly working) General Lighting is controlled in the same way (switch on and off together) so, when a General Lighting switch ON command is issued all the above power supply lines will be activated:

- Saloon lighting power supply line 1 (LV1) (if the full light mode is selected)
- Saloon lighting power supply line 2 (LV3) (if the full light mode is selected)
- Emergency lighting power supply line 1 (LV2-1)
- Emergency lighting power supply line 2 (LV2-2)
- Aesthetical & vestibules lighting power supply line (LV5)

The General Lighting ON/OFF commands can be issued by the staff acting on suitable devices located in the driver cabs, in the crew cabin and in all coaches.
In addition to that, also the TCMS can control the lighting system in some particular condition (i.e. desk activation, tunnel mode, load shedding mode).
From the crew cabin monitor it is possible to set the full light or reduced light mode. When in a coach the reduced lighting mode is selected, in that coach LV1 and LV3 will remain switched off.
Excluding the emergency lighting lines (LV2-1 and LV2-2) all the other lighting power supply lines are handled by the TCMS that controls them according to the commands issue by the staff/driver or to the operational condition of the train.

3.1.1.1.1 Normal condition

In normal condition the lighting system is switched on during the start-up phase of the vehicle (when the desk is activated).
In addition, suitable commands are foreseen to control the lighting system.
The saloon lighting level can be configured from the crew staff area monitor. Two configurations are foreseen: full lighting and reduced lighting

3.1.1.1.2 Failure cases

a) Fault on the LV1 line. Reduction of the medium lighting level.
b) Fault on the LV3 line. Reduction of the medium lighting level.
c) Fault on the LV2-1 line. Reduction of the medium lighting level.
d) Fault on the LV2-2 line. Reduction of the medium lighting level.
e) Fault on the LV5 line. Loss of aesthetical lighting.

3.1.1.1.3 Failure detection

a) LV1 line is continuously monitored by the TCMS.
b) LV3 line is continuously monitored by the TCMS.
c) LV2-1 line is continuously monitored by the TCMS.
d) LV2-2 line is continuously monitored by the TCMS.
e) LV5 line is continuously monitored by the TCMS.

3.1.1.1.4 Failure handling

a) No action for the driver and crew. The failure has to be repaired during maintenance.
b) No action for the driver and crew. The failure has to be repaired during maintenance.
c) No action for the driver and crew. The failure has to be repaired during maintenance.
d) No action for the driver and crew. The failure has to be repaired during maintenance.
e) No action for the driver and crew. The failure has to be repaired during maintenance.

3.1.1.2 Requirements

ID	2F_08.01.Zefiro-Europe.TRS.109
Description	It shall be possible to change the general lighting on train captain's command in all the defined operational modes (normal and dimmed - included "switched off").
Safety level	SIL 0

Nr	Description	Type	Derived to	User Interface	SIL
2F_08.01.Zefiro-Europe.CONCEPT.317	In the crew monitor it shall be foreseen a suitable page where the crew can select the following saloon lighting mode: - Full light - Reduced light	Derived Requirement	4S_09.03.05.-.TCMS Software	IDU	SIL 0
2F_08.01.Zefiro-Europe.CONCEPT.318	It shall be possible to set the saloon lighting mode in each single coach.	Derived Requirement	4S_09.03.05.-.TCMS Software	IDU	SIL 0
2F_08.01.Zefiro-Europe.CONCEPT.319	The saloon lighting mode selection shall remain stored in the TCMS so that even in case the train is switched off, at the next start up the saloon lighting configuration shall be the same as before.	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 0

ID	2F_08.01.Zefiro-Europe.TRS.11
Description	General Lighting shall start automatically when a cab is activated and Battery Charging is active.
Safety level	N/A

Nr	Description	Type	Derived to	User Interface	SIL
2F_08.01.Zefiro-Europe.CONCEPT.320	At the start up phase, once the battery charging is active, the TCMS shall activate the following group of lines for all the coaches in which Reduced light mode is active: · Emergency lighting line 1 (LV2-1) · Emergency lighting line 2 (LV2-2) · Aesthetical & vestibule lighting line (LV5)	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 0
2F_08.01.Zefiro-Europe.CONCEPT.321	At the start up phase, once the battery charging is active, the TCMS shall activate the following group of lines for all the coaches in which Full light mode is active: - Emergency lighting line 1 (LV2-1) · Emergency lighting line 2 (LV2-2) · Aesthetical & vestibule lighting line (LV5) · Normal lighting line 1 (LV1) · Normal lighting line 2 (LV3)	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 0
2F_08.01.Zefiro-Europe.CONCEPT.330	If the battery charging is active, when a cab is activated the TCMS shall activate the following group of lines for all the coaches in which Reduced light mode is active: · Emergency lighting line 1 (LV2-1)	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 0

	· Emergency lighting line 2 (LV2-2) · Aesthetical & vestibule lighting line (LV5)				
2F_08.01.Zefiro-Europe.CONCEPT.331	If the battery charging is active, when a cab is activated the TCMS shall activate the following group of lines for all the coaches in which Full light mode is active: - Emergency lighting line 1 (LV2-1) · Emergency lighting line 2 (LV2-2) · Aesthetical & vestibule lighting line (LV5) · Normal lighting line 1 (LV1) · Normal lighting line 2 (LV3)	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 0

ID	2F_08.01.Zefiro-Europe.TRS.67
Description	General Lighting shall remain in the current condition when cab is deactivated
Safety level	N/A

Nr	Description	Type	Derived to	User Interface	SIL
2F_08.01.Zefiro-Europe.CONCEPT.59	When a cab is deactivated the General Lighting shall remain in its current condition.	Derived Requirement	4S_09.03.05.-.TCMS Software		N/A

ID	2F_08.01.Zefiro-Europe.TRS.66
Description	It shall be possible to switch OFF/ON General Lighting from driver and from crew cabin and from all coaches.
Safety level	N/A

Nr	Description	Type	Derived to	User Interface	SIL
2F_08.01.Zefiro-Europe.CONCEPT.60	In order to control the General Lighting system, in each driver cab and in the crew cabin and in all coaches, a suitable control device shall be foreseen	Derived Requirement	4S_09.02.09.-.Conventional Train Control - Circuits	HW HMI	N/A
2F_08.01.Zefiro-Europe.CONCEPT.147	It shall be possible to switch ON the General Lighting from a cab and switch OFF it from the other cab or crew cabin or any coach, and viceversa also when trains are coupled.	Derived Requirement	4S_09.03.05.-.TCMS Software	HW HMI	SIL 0
2F_08.01.Zefiro-Europe.CONCEPT.162	It shall be possible to switch ON the General Lighting from a cab and switch OFF it from the other cab or crew cabin or any coach, and viceversa also when trains are coupled.	Derived Requirement	4S_09.02.09.-.Conventional Train Control - Circuits	HW HMI	SIL 0
2F_08.01.Zefiro-Europe.CONCEPT.322	If the battery charging is active and lighting system is switched off (by General Lighting switch OFF command), when a General Lighting switch ON command is issued, the TCMS shall set in all coachesthe lighting groups in accordance with the stored configuration	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 0
2F_08.01.Zefiro-Europe.CONCEPT.323	If the the lighting system is switched on (saloon lights set according to the crew area monitor) and at least one coach is set in “Reduced Light” mode, if a General Lighting switch ON command is issued, in all the coaches the TCMS shall activate the following lighting groups: · Normal lighting line 1 (LV1) · Normal lighting line 2 (LV3)	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 0
2F_08.01.Zefiro-Europe.CONCEPT.332	Starting from the “Full light” mode commanded in all coaches, if a new General Lighting switch ON command is issued, the TCMS shall de-activate the following lighting groups in the coaches where “Reduced light” on the crew area monitor is selected: · Normal lighting line 1 (LV1) · Normal lighting line 2 (LV3)	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 0
2F_08.01.Zefiro-Europe.CONCEPT.	When a General Lighting switch ON command is issued, in all coaches the following lighting groups	Derived Requirement	4S_09.02.09.-.Conventional Train		SIL 1

155	shall be activated: · Emergency lighting line 1 (LV2-1) · Emergency lighting line 2 (LV2-2)		Control - Circuits		
2F_08.01.Zefiro-Europe.CONCEPT.154	When a General Lighting switch ON command is issued, in all coaches the following lighting groups shall be activated: · Emergency lighting line 1 (LV2-1) · Emergency lighting line 2 (LV2-2)	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 0
2F_08.01.Zefiro-Europe.CONCEPT.153	When a General Lighting switch OFF command is issued, in all coaches the following lighting groups shall be deactivated: · Normal lighting line 1 (LV1) · Normal lighting line 2 (LV3) · Aesthetical lighting line (LV5)	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 0
2F_08.01.Zefiro-Europe.CONCEPT.152	When a General Lighting switch OFF command is issued, in all coaches the following lighting groups shall be deactivated: · Emergency lighting line 1 (LV2-1) · Emergency lighting line 2 (LV2-2)	Derived Requirement	4S_09.02.09.-.Conventional Train Control - Circuits		SIL 1
2F_08.01.Zefiro-Europe.CONCEPT.151	Status of the following lines shall be sent to TCMS: · Normal lighting line 1 (LV1) · Normal lighting line 2 (LV3) · Emergency lighting line 1 (LV2-1) · Emergency lighting line 2 (LV2-2) · Aesthetical lighting line (LV5)	Derived Requirement	4S_09.02.09.-.Conventional Train Control - Circuits		N/A
2F_08.01.Zefiro-Europe.CONCEPT.160	The TCMS shall monitor the current status of the following lines and show them on a suitable IDU page: · Normal lighting line 1 (LV1) · Normal lighting line 2 (LV3) · Emergency lighting line 1 (LV2-1) · Emergency lighting line 2 (LV2-2) · Aesthetical lighting line (LV5)	Derived Requirement	4S_09.03.05.-.TCMS Software	IDU	SIL 0
2F_08.01.Zefiro-Europe.CONCEPT.159	A failure on the Aesthetical lighting line shall not involve the other lines.	Derived Requirement	4S_09.02.09.-.Conventional Train Control - Circuits		N/A
2F_08.01.Zefiro-Europe.CONCEPT.158	A failure on the Normal lighting line 1 shall not involve the other lines.	Derived Requirement	4S_09.02.09.-.Conventional Train Control - Circuits		N/A
2F_08.01.Zefiro-Europe.CONCEPT.150	A failure on the Normal lighting line 2 shall not involve the other lines.	Derived Requirement	4S_09.02.09.-.Conventional Train Control - Circuits		N/A
2F_08.01.Zefiro-Europe.CONCEPT.324	If the battery charging is active and the General Lighting is active and the “full light” mode is set the TCMS shall store an event (Maintenance) if the Normal lighting line 1 (LV1) is not activated (Normal_Light1_Fault).	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	SIL 0
2F_08.01.Zefiro-Europe.CONCEPT.325	If the battery charging is active and the General Lighting is active and the “full light” mode is set the TCMS shall store an event (Maintenance) if the Normal lighting line 2 (LV3) is not activated (Normal_Light2_Fault).	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	SIL 0
2F_08.01.Zefiro-Europe.CONCEPT.156	If the battery charging is active and the General Lighting is active the TCMS shall store an event (Maintenance) if the Aesthetical lighting line (LV5) is not activated (Aesth_Light_Fault).	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	SIL 0
2F_08.01.Zefiro-Europe.CONCEPT.308	If the General Lighting is active the TCMS shall store an event (Maintenance) if the Emergency lightining line (LV2-1) is not activated (EM_Light1_Fault).	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	SIL 0
2F_08.01.Zefiro-Europe.CONCEPT.309	If the General Lighting is active the TCMS shall store an event (Maintenance) if the Emergency lightining line (LV2-2) is not activated (EM_Light2_Fault).	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	SIL 0
2F_08.01.Zefiro-Europe.CONCEPT.	In case of two coupled units, if general lights are turned ON in any of the units before coupling it	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 0

161	shall be turned ON in both units after coupling.				

3.1.2 Tunnel Mode

3.1.2.1 Design description

Every time the “Tunnel mode” operation is set, a General Lighting ON command is generated. When the “Tunnel mode” operation is reset the lighting system will remain in the current status (e.g. lighted).

3.1.2.1.1 Normal condition

In normal condition the lighting system is switched on during the start up phase of the vehicle (when the desk is activated). The lighting tunnel mode is a back-up of a possible bad operation of the staff ,that can order a switch off command of the lighting system, in order to prevent the train to enter a tunnel with the internal lighting switched off.

3.1.2.2 Requirements

ID	2F_08.01.Zefiro-Europe.TRS.15
Description	General Lighting shall be activated in the train whenever the “tunnel mode” is active, even if General Lighting was switched OFF by driver or crew
Safety level	N/A

Nr	Description	Type	Derived to	User Interface	SIL
2F_08.01.Zefiro-Europe.CONCEPT.326	When the “Tunnel Mode” operation is activated a General Lighting ON command shall be issued, that means that the following lighting groups shall be activated in each coach: · Emergency lighting line 1 (LV2-1) · Emergency lighting line 2 (LV2-2) · Aesthetical lighting line (LV5)	Derived Requirement	4S_09.03.05.-.TCMS Software		N/A
2F_08.01.Zefiro-Europe.CONCEPT.327	When the “Tunnel Mode” operation is activated in the coaches where the “Full light” mode is set, the TCMS shall activate the following lighting groups: · Normal lighting line 1 (LV1) · Normal lighting line 2 (LV3)	Derived Requirement	4S_09.03.05.-.TCMS Software		N/A

3.1.3 Emergency Lighting

3.1.3.1 Design description

After a preset time from the main power supply fault (e.g. low voltage system fed by the battery system), the internal lighting system emergency configuration is set by TCMS following a specific lighting groups disconnection sequence. When the emergency configuration is set, spot lights emission is opportunely reduced in order to contain the power consumption and at same time to respect the minimum lighting level required by the EN 13272.

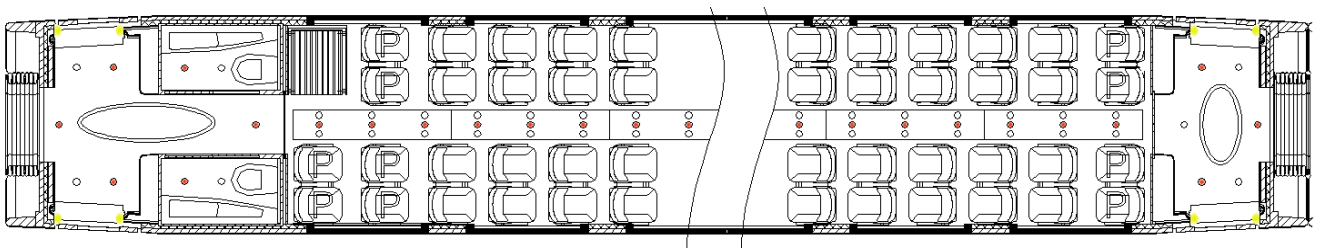


Figure 3 –Emergency lighting configuration (red and yellow spots are switched on)

The emergency lighting power supply line are derived directly from the batteries and split in two independent lines (LV2-1 and LV2-2).

3.1.3.1.1 Normal condition

In normal condition the emergency lighting system is switched on together with the general lighting during the start up phase of the vehicle (when the desk is activated).

3.1.3.1.2 Failure cases

a) Fault on the LV2-1 line. Loss of a redundancy (reduction of the medium lighting level)
b) Fault on the LV2-2 line. Loss of a redundancy (reduction of the medium lighting level)
c) Emergency lighting switch ON TL failure. Loss of HW ON command. The emergency lighting is switched on by the TCMS
d) Emergency lighting switch OFF TL failure. Loss of emergency lighting OFF command. The emergency lighting can not be switched off

3.1.3.1.3 Failure detection

a) LV2-1 line is continuously monitored by the TCMS.
b) LV2-2 line is continuously monitored by the TCMS.
c) LV2-1 and LV2-2 lines are continuously monitored by the TCMS.
d) LV2-1 and LV2-2 lines are continuously monitored by the TCMS.

3.1.3.1.4 Failure handling

a) No action for the driver and crew. The failure has to be repaired during maintenance.
b) No action for the driver and crew. The failure has to be repaired during maintenance.
c) No action for the driver and crew. The failure has to be repaired during maintenance.
d) No action for the driver and crew. The failure has to be repaired during maintenance.

3.1.3.2 Requirements

ID	2F_08.01.Zefiro-Europe.TRS.26
Description	Emergency lighting shall be powered directly from the battery
Safety level	N/A

Nr	Description	Type	Derived to	User Interface	SIL
2F_08.01.Zefiro-Europe.CONCEPT.62	Emergency Lighting circuitry of each coach shall be divided in two independent lines	Derived Requirement	4S_08.05.01.-.Interior Lights Systems		N/A
2F_08.01.Zefiro-Europe.CONCEPT.184	Emergency Lighting power supply of each coach shall be divided in two independent lines.	Derived Requirement	4S_09.02.09.-.Conventional Train Control - Circuits		N/A
2F_08.01.Zefiro-Europe.CONCEPT.183	Emergency Lighting power supply lines shall be derived direclty from the batteries (e.g. before the main battery contactor).	Derived Requirement	4S_09.02.09.-.Conventional Train Control - Circuits		N/A
2F_08.01.Zefiro-Europe.CONCEPT.182	A failure on an Emergency Lighting line shall not involve the other line.	Derived Requirement	4S_09.02.09.-.Conventional Train Control - Circuits		N/A

ID	2F_08.01.Zefiro-Europe.TRS.23
Description	Emergency Lighting shall be active while "general lighting" is active (as part of lighting).
Safety level	SIL 1

Nr	Description	Type	Derived to	User Interface	SIL
2F_08.01.Zefiro-Europe.CONCEPT.63	The Emergency Lighting switch ON command shall be transmitted along the train by means of a dedicated hardwired line (Emergency Lighting	Derived Requirement	4S_09.02.09.-.Conventional Train Control - Circuits		N/A

	switch ON train line). This is valid also when trains are coupled.				
2F_08.01.Zefiro-Europe.CONCEPT.185	The Emergency Lighting switch OFF command shall be transmitted along the train by means of a dedicated hardwired line (Emergency Lighting switch OFF train line) This is valid also when trains are coupled.	Derived Requirement	4S_09.02.09.-.Conventional Train Control - Circuits		N/A
2F_08.01.Zefiro-Europe.CONCEPT.186	The Emergency Lighting switch ON train line shall be active when the General Lighting switch ON command, issued from the cabs or crew cabin or any coach, is active.	Derived Requirement	4S_09.02.09.-.Conventional Train Control - Circuits		SIL 1
2F_08.01.Zefiro-Europe.CONCEPT.187	When a General Lighting switch ON command is issued from the cabs or crew cabin or any coach, in all coaches the following lighting groups shall be activated: · Emergency lighting line 1 (LV2-1) · Emergency lighting line 2 (LV2-2)	Derived Requirement	4S_09.02.09.-.Conventional Train Control - Circuits		SIL 1
2F_08.01.Zefiro-Europe.CONCEPT.190	Once activated, the Emergency Lighting lines of each coach shall remain actve as long as the General Lighting OFF command is not issued.	Derived Requirement	4S_09.02.09.-.Conventional Train Control - Circuits		SIL 1
2F_08.01.Zefiro-Europe.CONCEPT.189	The Emergency Lighting switch OFF train line shall be active when the General Lighting switch OFF command, issued from the cabs or crew cabin or any coach, is active.	Derived Requirement	4S_09.02.09.-.Conventional Train Control - Circuits		SIL 1
2F_08.01.Zefiro-Europe.CONCEPT.188	When a General Lighting switch OFF command is issued from the cabs or crew cabin or any coach, in all coaches the following lighting groups shall be deactivated: · Emergency lighting line 1 (LV2-1) · Emergency lighting line 2 (LV2-2)	Derived Requirement	4S_09.02.09.-.Conventional Train Control - Circuits		SIL 1

The TCMS will not be able to activate the Emergency Lighting switch OFF train line

ID	2F_08.01.Zefiro-Europe.TRS.68
Description	If battery charging is disabled general lights shall be turned off with a time delay and emergency light shall remain on.
Safety level	SIL 1

Nr	Description	Type	Derived to	User Interface	SIL
2F_08.01.Zefiro-Europe.CONCEPT.328	In case of battery charger system failure (e.g low voltage system fed by battery system) 30 minutes after that failure the following lighting groups shall be switched off in all coaches: · Normal lighting line 1 (LV1) · Normal lighting line 2 (LV3) · Aesthetical lighting line (LV5) · Safety lamp power supply line (LV4-1)	Derived Requirement	4S_09.02.09.-.Conventional Train Control - Circuits		SIL 0
2F_08.01.Zefiro-Europe.CONCEPT.329	In case of a battery charger system failure (e.g. low voltage system fed by the battery system), when the TCMS disconnects the following lighting groups: · Normal lighting line 1 (LV1) · Normal lighting line 2 (LV3) · Aesthetical lighting line (LV5) All the LED spot lights connected to the Emergency Lighting lines shall be set in to the “low power consuption” mode.	Derived Requirement	4S_08.05.01.-.Interior Lights Systems		SIL 1
2F_08.01.Zefiro-Europe.CONCEPT.194	In case of a battery charger system failure (e.g. low voltage system fed by the battery system), the emergency lighting system shall remain activated as long as the General Lighting OFF is not commanded by the staff.	Derived Requirement	4S_09.02.09.-.Conventional Train Control - Circuits		SIL 1

3.1.4 Safety lamps

3.1.4.1 Design description

The “safety lamp” function is provided by the first and last three emergency lights of the passenger saloon as shown in the figure below.

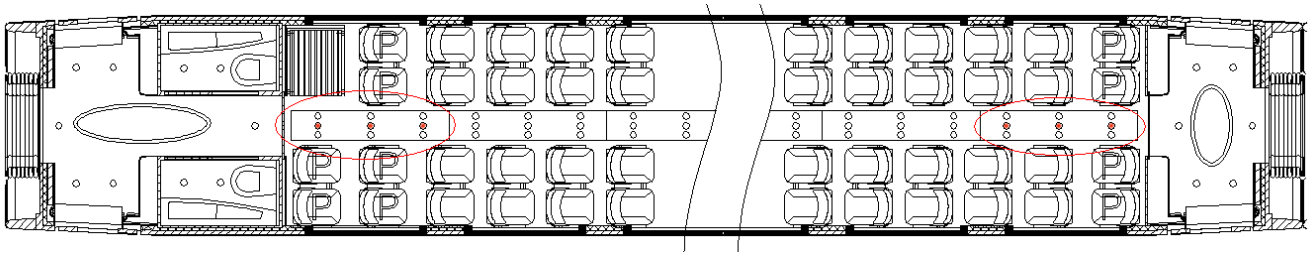


Figure 4 Safety lamps

Each of these two groups of lamps is provided with a dedicated battery.
The “safety lamp” function will be set as follow:
- automatically if the main batteries voltage goes under a preset value when the emergency lighting is switched on
- manually by a local (at coach level) or general command (at train level)

Normal operation (e.g. low voltage system of the train fed by the battery charger system) is used to recharge, or keep charged, the “safety lamp” internal batteries.

3.1.4.1.1 Normal condition

Normally the safety lamps are switched on in automatic way. The conditions that lead to the activation of the safety lamp are handled by each safety lamp control unit. The manual operation is used as a back up of the automatic mode and for the testing activities.

3.1.4.1.2 Failure cases

a) Safety lamp battery failure. Loss of the relevant safety lamp. Reduction of the medium lighting level.
b) Safety lamp control unit failure. Loss of the relevant safety lamp. Reduction of the medium lighting level.
c) Control train line failure. Loss of general commands (train level)

3.1.4.1.3 Failure detection

a) The status of the safety lamp battery is continuously monitored by the safety lamp control unit. In case of fault a message is sent to the TCMS
b) In case of a fault a message is sent to the TCMS.
c) By inspection during the periodic check done by the maintenance staff.

3.1.4.1.4 Failure handling

a) No action for the driver and crew. Failed safety lamp battery has to be changed during maintenance.
b) No action for the driver and crew. Failed safety lamp control unit has to be changed during maintenance.
c) If needed, the crew can switch on/off the safety lamps using the local control device (coach by coach). The faulty line has to be repaired during maintenance.

3.1.4.2 Requirements

ID	2F_08.01.Zefiro-Europe.TRS.86
Description	Safety lamps shall be activated automatically in case of the battery discharging if the system is in emergency lighting mode
Safety level	SIL 0

Nr	Description	Type	Derived to	User Interface	SIL
2F_08.01.Zefiro-Europe.CONCEPT.83	The “safety lamp” shall be switched on when the following conditions are verified: · the emergency lighting is switched on (i.e LV2-1 or LV2-2 are set) And · Safety lamp power supply line (LV4-1) is deactivated And · A voltage level below a preset value (TBD) of the emergency lighting lines is detected by the safety lamp control unit.	Derived Requirement	4S_08.05.01.-.Interior Lights Systems		SIL 0
2F_08.01.Zefiro-Europe.CONCEPT.207	The “safety lamp” shall be switched off when the normal condition (e.g. battery charger system active) restored, that means that: · Safety lamp power supply line (LV4-1) is activated And · Emergency lighting line voltage level is in the normal range (-25% / +30%)	Derived Requirement	4S_08.05.01.-.Interior Lights Systems		SIL 0

ID	2F_08.01.Zefiro-Europe.TRS.87
Description	Safety lamps shall be possible to activate manually in case of battery power supply is cut off. The manual activation shall be possible locally for each lamp and globally at train level
Safety level	N/A

Nr	Description	Type	Derived to	User Interface	SIL
2F_08.01.Zefiro-Europe.CONCEPT.85	It shall be possible to switch ON/OFF the safety lamps at coach level, acting on a dedicated device.	Derived Requirement	4S_08.05.01.-.Interior Lights Systems	HW HMI	N/A
2F_08.01.Zefiro-Europe.CONCEPT.209	It shall be possible to switch ON/OFF the safety lamps at train level (also when trains are coupled) from each coach, acting on a dedicated device.	Derived Requirement	4S_08.05.01.-.Interior Lights Systems	HW HMI	N/A
2F_08.01.Zefiro-Europe.CONCEPT.210	It shall be possible to switch ON/OFF the safety lamps also in case the train battery system is cut off.	Derived Requirement	4S_08.05.01.-.Interior Lights Systems		N/A
2F_08.01.Zefiro-Europe.CONCEPT.211	The Safety Lamps switch ON/OFF command shall be transmitted along the train by means of a dedicated hardwired line (Safety Lamps switch ON/OFF train line (LV4-2))	Derived Requirement	4S_09.02.09.-.Conventional Train Control - Circuits		N/A

The Safety Lamps switch ON/OFF train line (LV4-2) is commanded and acquired by the safety lamps kit without any contribution of the TCMS.

ID	2F_08.01.Zefiro-Europe.TRS.89
Description	Safety lamps shall in normal efficiency conditions be turned off. The internal battery shall then automatically recharge.
Safety level	SIL 0

Nr	Description	Type	Derived to	User Interface	SIL
2F_08.01.Zefiro-Europe.CONCEPT.87	In normal efficiency conditions of the train (e.g. battery charger system active), the Safety lamp power supply line (LV4-1) of each coach shall be activated.	Derived Requirement	4S_09.03.05.-.TCMS Software		N/A

The Safety lamp power supply line (LV4-1) is normally always powered and will be deactivated only in case of battery charger failure, according to the sequence described in the req. TRS 68.

Nr	Description	Type	Derived to	User Interface	SIL
2F_08.01.Zefiro-Europe.CONCEPT.214	When the Safety lamp power supply line (LV4-1) is activated, each safety lamp kit shall be set in the “charging mode” status indipendently by the condition of the Emergency lighting lines.	Derived Requirement	4S_08.05.01.-.Interior Lights Systems		N/A
2F_08.01.Zefiro-Europe.CONCEPT.217	Each safety lamp kit shall perform a monitoring of the internal batteries.	Derived Requirement	4S_08.05.01.-.Interior Lights Systems		N/A
2F_08.01.Zefiro-Europe.CONCEPT.216	The correct status of the internal batteries pack of each safety lamp kit shall be checked and sent to the TCMS.	Derived Requirement	4S_08.05.01.-.Interior Lights Systems		N/A
2F_08.01.Zefiro-Europe.CONCEPT.215	The status of battery of each Safety lamp kit shall be transmitted to the TCMS by means of a dedicated hardwired line (LV4-3 and LV4-4).	Derived Requirement	4S_09.02.09.-.Conventional Train Control - Circuits		N/A
2F_08.01.Zefiro-Europe.CONCEPT.218	When the power supply line LV4-1 is active, the TCMS shall store an event (Maintenance) in case of deactivation of the “Safety lamp xx - Battery status” signal	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	N/A

3.1.5 Reading Lamps

3.1.5.1 Design description

Reading lamps of the seats are controllable by the passengers that can control them acting on a push button. A test lamp function is foreseen in order to check the status of each reading lamp during maintenance activity.

3.1.5.1.1 Normal condition

See previous chapter.

3.1.5.1.2 Failure cases

a) Reading lamp failure. Loss of reading light and individual reading lamp function.

3.1.5.1.3 Failure detection

a) By inspection during the periodic check done by the maintenance staff.

3.1.5.1.4 Failure handling

a) No action for the driver and crew. Failed lamp has to be changed during maintenance.

3.1.5.2 Requirements

ID	2F_08.01.Zefiro-Europe.TRS.28
Description	Reading lights shall be enabled when the train is active, except when the train is in loadshedding mode.
Safety level	N/A

Nr	Description	Type	Derived to	User Interface	SIL
2F_08.01.Zefiro-Europe.CONCEPT.65	If the battery charging is active in all coaches the Reading Lights power supply line shall be activated.	Derived Requirement	4S_09.03.05.-.TCMS Software		N/A
2F_08.01.Zefiro-Europe.CONCEPT.	The Reading Lights power supply line shall be activated when ordered by TCMS	Derived Requirement	4S_09.02.09.-.Conventional Train		N/A

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ID	2F_08.01.Zefiro-Europe.TRS.69
Description	The individual reading lights ON/OFF shall be controlled by the passenger via a switch located at his seat.
Safety level	N/A

Nr	Description	Type	Derived to	User Interface	SIL
2F_08.01.Zefiro-Europe.CONCEPT.66	In each seat, a push button to control the reading light shall be foreseen.	Derived Requirement	4S_08.08.-.Seats		N/A
2F_08.01.Zefiro-Europe.CONCEPT.231	In order to control the reading light a set/reset logic shall be used.	Derived Requirement	4S_08.08.-.Seats		N/A
2F_08.01.Zefiro-Europe.CONCEPT.230	When the power supply coming from the train is removed the reading light logic shall be reset, that means that at the power up of the system the reading light is always switched off.	Derived Requirement	4S_08.08.-.Seats		N/A

ID	2F_08.01.Zefiro-Europe.TRS.77
Description	It shall be possible to turn on all reading lights in the train with one central command, for testing.
Safety level	N/A

Nr	Description	Type	Derived to	User Interface	SIL
2F_08.01.Zefiro-Europe.CONCEPT.67	It shall be possible to set the reading lights test in all coaches from the IDUs.	Derived Requirement	4S_09.03.05.-.TCMS Software	IDU	N/A
2F_08.01.Zefiro-Europe.CONCEPT.233	When the reading lights test is set from IDU, the reading lights test line of all coaches shall be energized for 5 minutes (final value TBD).	Derived Requirement	4S_09.03.05.-.TCMS Software		N/A
2F_08.01.Zefiro-Europe.CONCEPT.232	When the reading lights test is set from the train, the reading light of each seat shall remain switched on as long as the reading lights test is activated.	Derived Requirement	4S_08.08.-.Seats		N/A
2F_08.01.Zefiro-Europe.CONCEPT.288	The Reading Lights test line shall be activated when ordered by TCMS	Derived Requirement	4S_09.02.09.-.Conventional Train Control - Circuits		N/A

3.1.6 Toilet Lighting

3.1.6.1 Design description

The toilet lighting is always switched on when the toilet system is active. A part of the toilet lights are connected to the emergency lighting lines.

3.1.6.1.1 Normal condition

See previous chapter.

3.1.6.1.2 Failure cases

a) Single spot light failure. Reduction of the medium lighting level.

3.1.6.1.3 Failure detection

a) By inspection during the periodic check done by the maintenance staff.

3.1.6.1.4 Failure handling

a) No action for the driver and crew. Failed spot light has to be changed during maintenance.

3.1.6.2 Requirements

ID	2F_08.01.Zefiro-Europe.TRS.40
Description	Lavatory lighting shall be switched on when the toilet system is activated.
Safety level	N/A

Nr	Description	Type	Derived to	User Interface	SIL
2F_08.01.Zefiro-Europe.CONCEPT.69	At least a part of the toilet lighting system shall be switched on every time the toilet system is activated.	Derived Requirement	4S_08.03.-.Toilet Systems		N/A

With the emergency lighting is switched off not all of the toilet lighting is turned ON when the toilet system is activated

ID	2F_08.01.Zefiro-Europe.TRS.81
Description	There shall be two spotlights connected to “emergency lighting” in the toilet lighting system.
Safety level	SIL 1

Nr	Description	Type	Derived to	User Interface	SIL
2F_08.01.Zefiro-Europe.CONCEPT.89	At least two LED spot lights of the toilet lighting system shall be lit when emergency lighting is set.	Derived Requirement	4S_08.03.-.Toilet Systems		N/A
2F_08.01.Zefiro-Europe.CONCEPT.251	At least two LED spot lights of the toilet lighting system shall be connected to the emergency lighting line.	Derived Requirement	4S_09.02.09.-.Conventional Train Control - Circuits		SIL 1

3.1.7 Kitchen Lighting

3.1.7.1 Design description

3.1.7.1.1 Normal condition

See previous chapter.

3.1.7.1.2 Failure cases

a) Single spot light failure. Reduction of the medium lighting level.

3.1.7.1.3 Failure detection

a) By inspection during the periodic check done by the maintenance staff.

3.1.7.1.4 Failure handling

a) No action for the driver and crew. Failed spot light has to be changed during maintenance.

3.1.7.2 Requirements

ID	2F_08.01.Zefiro-Europe.TRS.44
Description	It shall be possible to switch ON/OFF the kitchen lighting by a switch located in the kitchen.
Safety level	N/A

Nr	Description	Type	Derived to	User Interface	SIL
2F_08.01.Zefiro-Europe.CONCEPT.70	It shall be possible to switch ON/OFF the kitchen lighting by a switch located in the kitchen.	Derived Requirement	4S_08.04.-.Catering	HW HMI	N/A

ID	2F_08.01.Zefiro-Europe.TRS.82
Description	The light for the working plate shall be possible to switch ON/OFF individually
Safety level	N/A

Nr	Description	Type	Derived to	User Interface	SIL
2F_08.01.Zefiro-Europe.CONCEPT.91	The light for the working plate shall be possible to switch ON/OFF individually	Derived Requirement	4S_08.04.-.Catering	HW HMI	N/A

ID	2F_08.01.Zefiro-Europe.TRS.83
Description	There shall be at least two spotlights connected to "emergency lighting" in the kitchen lighting system..
Safety level	SIL 1

Nr	Description	Type	Derived to	User Interface	SIL
2F_08.01.Zefiro-Europe.CONCEPT.92	At least two LED spot lights of the the kitchen lighting system shall be lit when emergency lighting is set.	Derived Requirement	4S_08.04.-.Catering		N/A
2F_08.01.Zefiro-Europe.CONCEPT.260	At least two LED spot lights of the the kitchen lighting system shall be lit when emergency lighting is set.	Derived Requirement	4S_09.02.09.-.Conventional Train Control - Circuits		SIL 1

3.1.8 Lighting in Crew Cabin

3.1.8.1 Function Design

3.1.8.1.1 Normal condition

See previous chapter.

3.1.8.1.2 Failure cases

a) Single spot light failure. Reduction of the medium lighting level.

3.1.8.1.3 Failure detection

a) By inspection during the periodic check done by the maintenance staff.

3.1.8.1.4 Failure handling

a) No action for the driver and crew. Failed spot light has to be changed during maintenance.

3.1.8.2 Requirements

ID	2F_08.01.Zefiro-Europe.TRS.48
Description	Lights in the crew cabin shall be connected at least to “emergency lighting”.
Safety level	SIL 1

Nr	Description	Type	Derived to	User Interface	SIL
2F_08.01.Zefiro-Europe.CONCEPT.71	At least one LED spot lights of the crew cabin shall be lit when emergency lighting is set.	Derived Requirement	4S_08.05.01.-.Interior Lights Systems		N/A
2F_08.01.Zefiro-Europe.CONCEPT.269	At least one LED spot lights of the crew cabin shall be lit when emergency lighting is set.	Derived Requirement	4S_09.02.09.-.Conventional Train Control - Circuits		SIL 1

ID	2F_08.01.Zefiro-Europe.TRS.70
Description	Lights in the crew cabin shall be controlled ON/OFF by a switch.
Safety level	N/A

Nr	Description	Type	Derived to	User Interface	SIL
2F_08.01.Zefiro-Europe.CONCEPT.72	It shall be possible to switch ON/OFF the crew cabin lights acting on dedicated device (switch selector).	Derived Requirement	4S_09.02.09.-.Conventional Train Control - Circuits	HW HMI	N/A

4 Signal Interface

4.1 I/O Interface

4.1.1 Supplier system

Inputs:

Nr	Functional name	Type	Linked to requirement
2F_08.01.Zefiro-Europe.CONCEPT.28	--		

Outputs:

Nr	Functional name	Type	Linked to requirement
2F_08.01.Zefiro-Europe.CONCEPT.29	Safety lamp 1 - Battery status OK		2F_08.01.Zefiro-Europe.CONCEPT.215
2F_08.01.Zefiro-Europe.CONCEPT.270	Safety lamp 2 - Battery status OK		2F_08.01.Zefiro-Europe.CONCEPT.215

4.1.2 TCMS SW

Inputs:

Nr	Functional name	Type	Linked to requirement
2F_08.01.Zefiro-Europe.CONCEPT.333	General lighting switch on command		2F_08.01.Zefiro-Europe.CONCEPT.320 2F_08.01.Zefiro-Europe.CONCEPT.321 2F_08.01.Zefiro-Europe.CONCEPT.326 2F_08.01.Zefiro-Europe.CONCEPT.327 2F_08.01.Zefiro-Europe.CONCEPT.322 2F_08.01.Zefiro-Europe.CONCEPT.323 2F_08.01.Zefiro-Europe.CONCEPT.153 2F_08.01.Zefiro-Europe.CONCEPT.328 2F_08.01.Zefiro-Europe.CONCEPT.329
2F_08.01.Zefiro-Europe.CONCEPT.334	General lighting switch off command		2F_08.01.Zefiro-Europe.CONCEPT.320 2F_08.01.Zefiro-Europe.CONCEPT.321 2F_08.01.Zefiro-Europe.CONCEPT.326 2F_08.01.Zefiro-Europe.CONCEPT.327 2F_08.01.Zefiro-Europe.CONCEPT.322 2F_08.01.Zefiro-Europe.CONCEPT.153 2F_08.01.Zefiro-Europe.CONCEPT.328 2F_08.01.Zefiro-Europe.CONCEPT.329
2F_08.01.Zefiro-Europe.CONCEPT.272	Normal lighting line 1 (LV1) status (ON feedback)		2F_08.01.Zefiro-Europe.CONCEPT.151
2F_08.01.Zefiro-Europe.CONCEPT.275	Normal lighting line 2 (LV3) status (ON feedback)		2F_08.01.Zefiro-Europe.CONCEPT.151
2F_08.01.Zefiro-Europe.CONCEPT.274	Emergency lighting line 1 (LV2-1) status (ON feedback)		2F_08.01.Zefiro-Europe.CONCEPT.151
2F_08.01.Zefiro-Europe.CONCEPT.273	Emergency lighting line 2 (LV2-2) status (ON feedback)		2F_08.01.Zefiro-Europe.CONCEPT.151
2F_08.01.Zefiro-Europe.CONCEPT.276	Aesthetical lighting line (LV5) status (ON feedback)		2F_08.01.Zefiro-Europe.CONCEPT.151
2F_08.01.Zefiro-Europe.CONCEPT.278	Safety lamp 1 - Battery status OK		2F_08.01.Zefiro-Europe.CONCEPT.216
2F_08.01.Zefiro-Europe.CONCEPT.	Safety lamp 2 - Battery status OK		2F_08.01.Zefiro-Europe.CONCEPT.216

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Outputs:

Nr	Functional name	Type	Linked to requirement
2F_08.01.Zefiro-Europe.CONCEPT.335	Normal lighting line 1 (LV1) contactor		2F_08.01.Zefiro-Europe.CONCEPT.320 2F_08.01.Zefiro-Europe.CONCEPT.321 2F_08.01.Zefiro-Europe.CONCEPT.326 2F_08.01.Zefiro-Europe.CONCEPT.327 2F_08.01.Zefiro-Europe.CONCEPT.322 2F_08.01.Zefiro-Europe.CONCEPT.153 2F_08.01.Zefiro-Europe.CONCEPT.328 2F_08.01.Zefiro-Europe.CONCEPT.329
2F_08.01.Zefiro-Europe.CONCEPT.336	Normal lighting line 2 (LV3) contactor		2F_08.01.Zefiro-Europe.CONCEPT.320 2F_08.01.Zefiro-Europe.CONCEPT.321 2F_08.01.Zefiro-Europe.CONCEPT.326 2F_08.01.Zefiro-Europe.CONCEPT.327 2F_08.01.Zefiro-Europe.CONCEPT.322 2F_08.01.Zefiro-Europe.CONCEPT.323 2F_08.01.Zefiro-Europe.CONCEPT.153 2F_08.01.Zefiro-Europe.CONCEPT.328 2F_08.01.Zefiro-Europe.CONCEPT.329
2F_08.01.Zefiro-Europe.CONCEPT.337	Emergency lighting line 1 (LV2-1) contactor		2F_08.01.Zefiro-Europe.CONCEPT.320 2F_08.01.Zefiro-Europe.CONCEPT.321 2F_08.01.Zefiro-Europe.CONCEPT.326 2F_08.01.Zefiro-Europe.CONCEPT.327 2F_08.01.Zefiro-Europe.CONCEPT.155 2F_08.01.Zefiro-Europe.CONCEPT.187
2F_08.01.Zefiro-Europe.CONCEPT.338	Emergency lighting line 2 (LV2-2) contactor		2F_08.01.Zefiro-Europe.CONCEPT.320 2F_08.01.Zefiro-Europe.CONCEPT.321 2F_08.01.Zefiro-Europe.CONCEPT.326 2F_08.01.Zefiro-Europe.CONCEPT.327 2F_08.01.Zefiro-Europe.CONCEPT.155 2F_08.01.Zefiro-Europe.CONCEPT.187
2F_08.01.Zefiro-Europe.CONCEPT.339	Aesthetical lighting line (LV5) contactor		2F_08.01.Zefiro-Europe.CONCEPT.320 2F_08.01.Zefiro-Europe.CONCEPT.321 2F_08.01.Zefiro-Europe.CONCEPT.326 2F_08.01.Zefiro-Europe.CONCEPT.327 2F_08.01.Zefiro-Europe.CONCEPT.328 2F_08.01.Zefiro-Europe.CONCEPT.329 2F_08.01.Zefiro-Europe.CONCEPT.322 2F_08.01.Zefiro-Europe.CONCEPT.323 2F_08.01.Zefiro-Europe.CONCEPT.153
2F_08.01.Zefiro-Europe.CONCEPT.284	Reading lights test		2F_08.01.Zefiro-Europe.CONCEPT.233
2F_08.01.Zefiro-Europe.CONCEPT.340	Safety lamp power supply line (LV4-1)		2F_08.01.Zefiro-Europe.CONCEPT.87 2F_08.01.Zefiro-Europe.CONCEPT.328 2F_08.01.Zefiro-Europe.CONCEPT.329

4.2 Bus Interface

Supplier to TCMS SW:

Nr	Functional name	Type	Linked to requirement
2F_08.01.Zefiro-Europe.CONCEPT.32			

TCMS SW to Supplier:

Nr	Functional name	Type	Linked to requirement
2F_08.01.Zefiro-Europe.CONCEPT.33			

5 References

[1] TRD LIGHTING SYSTEM EC01P006865B
[2] 3EST000225-7908 SFRS Battery
[3] EC01P008890B VZI50 internal lighting FRS